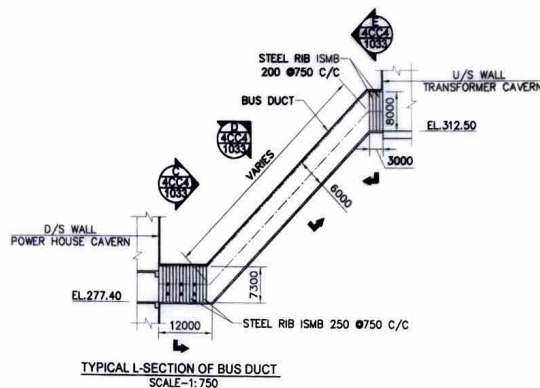
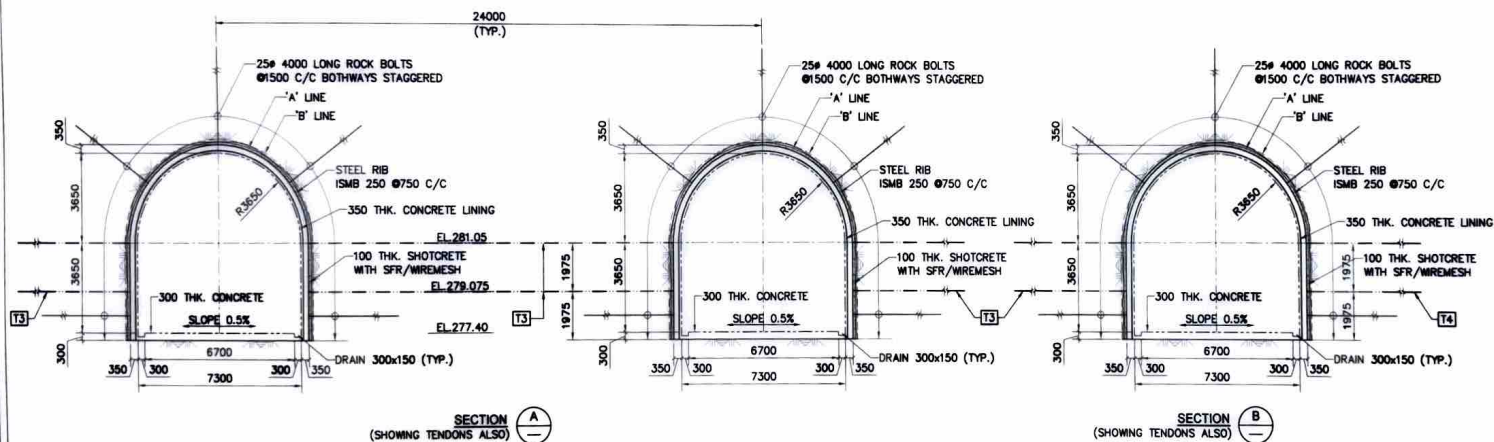
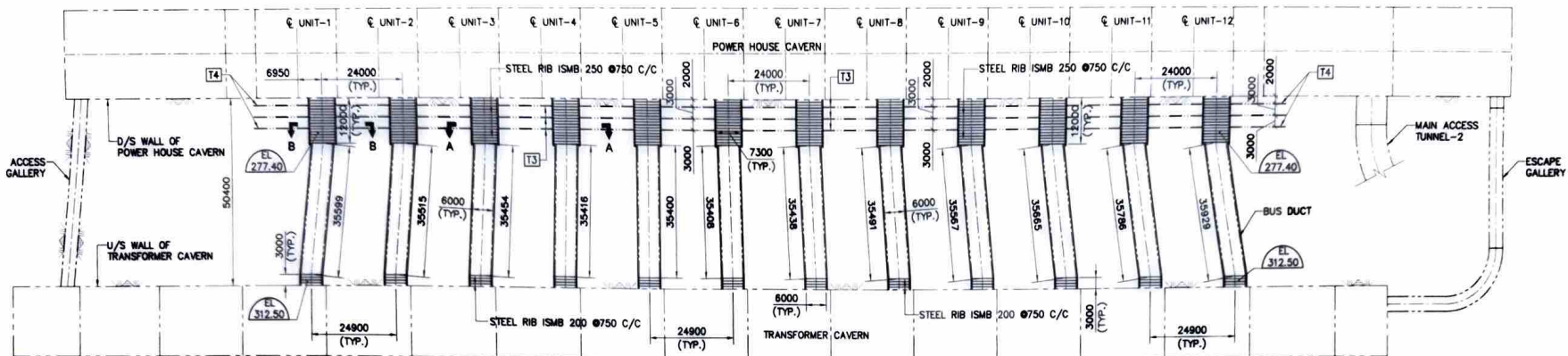




NOTES FOR POST TENSIONED ROCK BOLTS (TENDONS) :

1. TENDONS, THEIR INSTALLATION, TENSIONING AND TESTING SHALL CONFORM TO SECTION B.4 OF TECHNICAL SPECIFICATIONS.
2. THE ENTIRE WORK OF DRILLING, GROUTING, FABRICATION, INSTALLATION AND TESTING OF TENDONS SHALL BE CARRIED OUT BY MANUFACTURER'S AGENCY HAVING EXPERIENCE, EQUIPMENT AND TECHNOLOGY FOR EXECUTION OF THESE WORKS AND UNDER STRICT QUALITY CONTROL.
3. THE CONTRACTOR SHALL SUBMIT COMPLETE DETAILS CONCERNING TYPE AND MATERIALS OF THE TENDONS TOGETHER WITH MANUFACTURER'S INSTRUCTIONS AND CERTIFICATIONS, METHODS OF INSTALLATION, CORROSION PROTECTION, STRESSING EQUIPMENT, GROUTING METHOD AND EQUIPMENT, PROPOSAL FOR SUITABILITY AND STRESSING TESTS AND EQUIPMENT.
4. TYPES OF POST TENSIONED ROCK BOLTS (TENDONS): -
 - (a) type-1: CONSISTING OF HIGH STRENGTH STEEL TENDON FITTED WITH A STRESSING ANCHORAGE (FIXED LENGTH) AT THE INNER END, A MEANS PERMITTING FORCE TO BE TRANSFERRED, AND A HEAD ON THE ROCK SURFACE AT THE OUTER END.
 - (b) type-2: CONSISTING OF HIGH STRENGTH STEEL TENDON FITTED ON HEAD ON THE ROCK SURFACE AT BOTH END, WITH ONE END FOR ANCHORAGE, A MEANS PERMITTING FORCE TO BE TRANSFERRED, AND OTHER END FOR STRESSING.
5. CONSOLIDATION GROUTING, WHEREVER REQUIRED SHALL BE CARRIED OUT PRIOR TO INSTALLATION OF TENDONS.

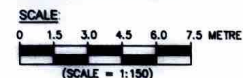
LEGENDS -

- T3** (TYPE-2)-600KN GUARANTEED ULTIMATE TENSILE STRENGTH (GUTS), LOCK OFF LOAD SHALL BE 60% OF GUTS, TOTAL LENGTH=16M.
- T4** (TYPE-1)-600KN GUARANTEED ULTIMATE TENSILE STRENGTH (GUTS), LOCK OFF LOAD SHALL BE 60% OF GUTS, TOTAL LENGTH=18M.

SL. NO.	TYPE	LENGTH (M)	NOS.	TOTAL LENGTH (TENTATIVE)
1.	T3	16	66	1056
2.	T4	18	12	216

REFERENCE DRAWINGS:-

1. NHDB-4AC4-41-DD-1003.
2. NHDB-4CC4-41-DD-0004 TO 0006
3. NHDB-4CC4-41-DD-1012 TO 1025.
4. NHDB-4CC4-41-DD-1028 TO 1030
5. NHDB-4GC4-41-DD-1005 & 1006.
6. NHDB-4GC4-41-DD-1009 TO 1013.



	एन एच पी सी लिमिटेड NHPC Limited (A Public Limited Company)
	विद्युत बहुउद्देशीय निगम DIBANG MULTIPURPOSE PROJECT
POWER HOUSE BUS DUCTS EXCAVATION, ROCK SUPPORTS AND CABLE TENDONS DETAILS	
(SHEET 1 OF 2)	
DESIGN JUN 2026	APPROVED DD 1032 00